

ARNAV MALHOTRA

Software Engineer | Distributed Systems & Infrastructure

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EDUCATION

Bachelor of Engineering in Software Engineering

Expected May 2027

York University

Toronto, ON

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Kuperman Investments

Remote

- Replaced the trading desk's manual spreadsheet workflow by designing a real-time market-data platform on **AWS (S3/Parquet, DuckDB)**, creating a single source of truth over **3.2M+ records** with automated freshness alerting
- Scaled collection to meet desk demand by engineering a fault-tolerant distributed system in **Go (100+ concurrent workers)**, feeding real-time data into pricing and trading models
- Eliminated silent data gaps that broke downstream models by adding deduplication, automated backfill, and dead-letter handling, guaranteeing correctness under partial node failures

Software Engineering Intern

Jul 2025 – Sep 2025

Quarkle.ai

Remote

- Cut median retrieval latency by **35%** on a real-time AI document-Q&A pipeline by adding websocket streaming and request-level caching to the **Python/FastAPI** backend
- Grew weekly shared-document usage **2.1x** across a **500+ user** beta by shipping publishing and collaboration workflows with permission controls and shareable public links

Software Development Contractor

Apr 2025 – Jun 2025

Markis Investments

Remote

- Delivered an end-to-end resort-management suite (**Next.js 15, React 19, TypeScript, Supabase/PostgreSQL**): a direct-booking site, guest portals, and an operations dashboard with month-over-month financial analytics (**Chart.js**) and **Hostaway API** reservation sync
- Architected a **multi-tenant QR-based event check-in system (7 role-based access tiers, 80+ API routes)** that cut average guest processing from **2 minutes** to under **20 seconds**

Research Fellow

May 2024 – Jul 2024

Indian Institute of Technology, Roorkee

Roorkee, India

- Achieved a **0.52 ± 0.05** Dice score on epilepsy lesion segmentation by training **nnU-Net** on MRI from **85 subjects** with 5-fold cross-validation; published in Frontiers in Artificial Intelligence

PROJECTS

Forge: Distributed Task Queue | Go, Raft, gRPC, BadgerDB, Prometheus, Docker

[GitHub](#)

- Building a fault-tolerant distributed job queue with a **Raft-replicated coordinator** (leader election + log replication), effectively-once delivery, retries, and dead-letter queues, validated by a chaos-test harness

OneTimeTable: AI Study Planner | Next.js, TypeScript, FastAPI, Redis, LLM

onetimetable.org

- Shipped a live SaaS that parses course syllabi with an **LLM** to auto-extract deadlines and exams, exporting them to **Google** and **Apple Calendar**; cut load times **60%** with **Redis** caching

TECHNICAL SKILLS

Languages: Java, Python, Go, C, JavaScript, TypeScript, SQL

Distributed & Infra: AWS (EC2, S3, Lambda, CloudWatch), gRPC, Raft, Docker, systemd, Linux, Parquet

Frameworks: FastAPI, Next.js, React, Node.js, Flask

Databases: PostgreSQL, DuckDB, Redis, MongoDB, SQLite

Libraries/Tools: PyTorch, NumPy, Pandas, Playwright, Git, Tailwind CSS